

Genome-wide identification and expression analysis of the B-box gene family in the Apple (*Malus domestica* Borkh.) genome

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Abstract The B-box proteins (BBXs) are a family of zinc finger proteins containing one/two B-box domain(s). Compared with intensive studies of animal BBXs, investigations of the plant BBX family are limited, though some specific plant BBXs have been demonstrated to act as transcription factors in the regulation of flowering and photomorphogenesis. In this study, using a global search of the apple (*Malus domestica* Borkh.) genome, a total of 64 members of BBX (*MdBBX*) were identified. All the *MdBBX*s were divided into five groups based on the phylogenetic relationship, numbers of B-boxes contained and whether there was with an additional CCT domain. According to the characteristics of organ-specific expression, *MdBBX*s were divided into three groups based on the microarray information. An analysis of *cis*-acting elements showed that elements related to the stress response were prevalent in the promoter sequences of most *MdBBX*s. Twelve *MdBBX* members from different groups were randomly selected and exposed to abiotic stresses. Their expressions were up-regulated to some extent in the roots and leaves. Six among 12 *MdBBX*s were sensitive to osmotic pressure, salt, cold stress and exogenous abscisic acid treatment, with their expressions enhanced

more than 20-fold. Our results suggested that *MdBBX*s may take part in response to abiotic stress.

Keywords B-box proteins · Stress response · *Cis*-acting elements · Organ-specific expression · Apple

Introduction

The B-box domain (B-box for short) is a conserved motif found in more than 1500 proteins of multicellular species and some unicellular eukaryotes, including green algae, moss, *Arabidopsis*, rice, banana, tomato and mouse (Meroni and Diez-Roux 2005; Crocco and Botto 2013; Gangappa and Botto 2014). Consisting of approximately 40 amino acids, the B-box is predicted to be involved in protein–protein interactions. In animals, the B-box is often associated with various domains forming different kinds of protein complex, such as RBCC/TRIM (for RING, B-box, coiled-coil/TRIPARTITE MOTIF) (Borden 1998; Torok and Etkin 2001). The RBCC/TRIM proteins play important roles in diverse cellular processes including apoptosis, regulation of the cell cycle and viral responses (Meroni and Diez-Roux 2005). In some RBCC/TRIM proteins, the B-box functions in recognition of the substrate of the RBCC/TRIM proteins (Torok and Etkin 2001; Meroni and Diez-Roux 2005). Moreover, the B-box is also involved in the regulation of transcription (Beenders et al. 2007).

Different from animal B-box proteins (BBXs), some plant BBXs have an additional CCT domain (for Constans, Constans-like and Timing of CAB1) (Strayer et al. 2000). Proteins bearing the CCT domain were previously named as COL proteins (for Constans and Constans-like). Lately, plant BBXs without the CCT domain were identified, therefore, B-box containing proteins with and without

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the CCT domain were renamed as BBXs as the uniform terminology (Khanna et al. 2009). As transcription factors, plant BBXs belongs to a subgroup of zinc finger proteins, which are involved in the regulation of transcription by binding with other proteins, instead of binding with DNA or RNA. Information about transcription factors in plants, including plant BBXs are updated in PlantTFDB 4.0 (<http://planttfdb.cbi.pku.edu.cn/index.php>) (Jin et al. 2017). The best-known *Arabidopsis* BBX protein, AtBBX1 (also called CO in the original publication), with two B-boxes and one CCT domain plays essential roles in the photoperiod regulation of flowering (Putterill et al. 1995; Robson et al. 2001).

The AtBBX1 deficient mutant of *Arabidopsis*, delayed flowering under a long light period, while over-expressing AtBBX1 in wild-type made the plants flower early both under long and short days (Putterill et al. 1995; Suarez-Lopez et al. 2001). Other *Arabidopsis* BBXs were also found to regulate flowering times, such as AtBBX4 and AtBBX7 (Cheng and Wang 2005; Datta et al. 2006). Some *Arabidopsis* BBXs are related to seedling de-etiolation, controlling hypocotyl growth, anthocyanin production and chlorophyll accumulation (Datta et al. 2008; Wang et al. 2013; Gangappa and Botto 2014). The Plant BBX genes were also related to flowering regulation and distinct circadian regulation in banana (Chaurasia et al. 2016). *Arabidopsis* BBX5 is recently reported to be involved in signaling pathways induced by abiotic and biotic stresses (Min et al. 2015).

Research on the plant BBX family has been performed in *Arabidopsis*, rice, tomato, pear and potato. Most studies focused on herbaceous plant species about the BBX evolutionary and functions in plant development (Khanna et al. 2009; Huang et al. 2012; Gangappa and Botto 2014; Chu et al. 2016; Talar et al. 2017; Zou et al. 2017).

As one of the most important economic crops, apple is widely cultivated in the temperate regions of the world. The complete apple (*Malus domestica* Borkh.) genome was released in 2010, offering the possibility of investigating the BBX gene family in the species (Velasco et al. 2010). Similar with *Arabidopsis* BBX, some apple BBX genes were reported to be involved in the regulation of flowering and anthocyanin biosynthesis (Jeong et al. 1999; Bai et al. 2014).

In this study, through a genome-wide survey, all apple BBXs (*MdBBXs*) were identified. Their chromosomal distributions, gene structures and expression patterns in different organs were investigated. Promoter analysis suggested that the elements related to stresses widely existed in that of most *MdBBXs*. Of 64 *MdBBXs*, 12 were randomly selected to check the expressions under different stresses. This study may provide a foundation for further functional studies of plant BBXs on responses and adaption to abiotic stresses.

Materials and methods

Identification of *BBX* encoding genes in the apple (*Malus domestica* Borkh.) genome

To identify the genes encoding BBXs in the apple (*Malus domestica* Borkh.) genome, two different methods were used. In the first method, the BLASTp program was used to search for potential BBX encoding genes in the whole apple (*Malus domestica* Borkh.) genome, using the known sequences of BBXs from different plants as queries, with the *E* value cutoff set at 0.001 as did in similar studies (Li et al. 2011; Jia et al. 2013; Cui et al. 2015). In the second method, the hidden Markov model profile (Pfam00643) of the B-box domain established by the Pfam database (<http://pfam.xfam.org/>) was used as a query to search for all of the annotated proteins in the complete apple (*Malus domestica* Borkh.) genome database (<http://www.rosaceae.org/>) using HMMER 3.0 software (Eddy 1998). Each potential candidate acquired by the above two methods was further examined the presence of B-box, by submitting them to InterPro (<http://www.ebi.ac.uk/interpro/>) and SMART (<http://smart.embl-heidelberg.de/>), respectively. The molecular weight and isoelectric point (pI) of *MdBBXs* identified were calculated by the ExPASy Compute pI/Mw Program (http://web.expasy.org/compute_pi/) (Wilkins et al. 1999).

Construction of *MdBBXs* phylogenetic tree and motif structure analysis

Multiple sequence alignments of the full-length *MdBBXs* were performed with the ClustalX program (Version 2.1; <http://www.clustal.org/>) (Larkin et al. 2007). Phylogenetic trees were constructed based on the protein sequences of the *MdBBXs* using the neighbor-joining method of the program MEGA5.0 (Tamura et al. 2011). The motif logos of the B-box domain of apple BBXs were generated by submitting the sequences to the MEME website (<http://meme.nbcr.net/meme/cgi-bin/meme.cgi>).

Chromosomal location of *MdBBXs* and gene structure analysis

Chromosomal location data of the *MdBBXs* identified above were retrieved from genome annotations downloaded from the Genome Database for *Rosaceae* (short for the GDR, <https://www.rosaceae.org/>) according to the gene ID. MapDraw software was used to draw the detailed location of *MdBBXs* on the corresponding chromosomes (Liu and Meng 2003). To generate the gene structure of *MdBBXs*, the online software of GSDS (gene structure display server, <http://gsds.cbi.pku.edu.cn/>) was used. Tandem duplications of adjacent genes were characterized, if their similarities

were more than 70% and located within the distance of ten predicted genes or within 30 kb (Du et al. 2013). To elucidate the evolutionship of *MdBBX*, the ratio of Ka/Ks was used to express the positive selection (with ratio greater than 1), negative or stabilizing selection (Ka/Ks less than 1), and neutral selection (Ka/Ks equal to 1), respectively. While the Ka (nonsynonymous substitutions) and Ks (synonymous substitutions) were calculated by software MEGA 5.0 (Doerks et al. 2002).

Analysis organ-specific expression pattern of *MdBBX* genes

The expression profile of apple (*Malus domestica* Borkh.), GSE42873, downloaded from NCBI GEO datasets (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE42873>) was used to investigate the organ-specific expression of *MdBBX* genes. The transcriptional profiling (GSE42873) included eight apple organs from different apple cultivars. More detailed information about the samples was also shown at above website. The gene names in GSE42873 were consistent with the names used above in the GDR database. Expression data for the identified *MdBBX* genes were obtained from the two datasets using their unigene IDs and gene names, respectively (Cui et al. 2015). The hierarchical clustering of *MdBBX*s was implemented using software Cluster 3.0 to make the heatmap and clustering tree.

Cis-acting elements searching in the promoters of *MdBBX*s

Approximately 1500 bp of genomic DNA sequence upstream of the transcriptional start sites (the promoter sequences of *MdBBX*s) were used to analyze the *cis*-acting elements. Each sequence was submitted to the PlantCARE database (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) to identify the presence of various *cis*-acting elements (Lescot et al. 2002).

Plant material, growth conditions and stress treatments to plants

The germinated apple seeds (*Golden Delicious*, *Malus domestica* Borkh.) with two green leaves were grown in the Hoagland solution at 25 °C under a photoperiod of 16 h light/8 h dark in climate chamber for about one year (Fig. S1). The seedlings with the average heights of 40 cm and 20 leaves were used for stress treatments (Zhao et al. 2012). The whole seedlings were used for exogenous ABA and chilling treatments, by spraying on leaves with 100 µM abscisic acid or put them at 4 °C. As to PEG and salt treatments, 20% polyethylene glycol 6000 (PEG-6000) or 250 mM NaCl were used to replace the Hoagland solution. The concentrations of

ABA, PEG and NaCl were determined based on the similar investigations (Zhao et al. 2012, 2016; Yuan et al. 2013; Xu et al. 2016). Similar with the reference, each sample was from three different seedlings and collected at 1, 3, 6, 9 and 12 h (Yuan et al. 2013). The samples of roots and leaves collected were stored at −70 °C after freezing in liquid nitrogen until for use.

RNA extraction and quantitative real-time PCR (qRT-PCR) analysis

The total RNA was extracted from the above material using the CTAB procedure (Gasic et al. 2004). The RNA quality (the ratio of A_{260}/A_{280}) and concentrations were determined using a NanoDrop Spectrometer (ND-1000 Spectrophotometer, Peqlab). Qualified RNA was used to synthesize cDNA with a cDNA synthesis kit (TransGen Biotech, Beijing, China). The *actin* gene from apple was used as an internal control for expression normalization in the qRT-PCR as similar investigations did and the relative expression levels were calculated by the method of $2^{-\Delta\Delta C_t}$ (Su et al. 2013; Xu et al. 2016). To ensure the samples obtained were credible, one apple stress-specific gene (*DREB* gene: MDP0000147009) was additionally selected as positive control, which expression has been demonstrated to be up-regulated under stress treatments (Zhao et al. 2012). The specific primers for amplifying *actin*, *DREB* and *MdBBX* genes were used for qRT-PCR and are listed in Table S1.

Results

Identification of the B-box containing proteins in *Malus domestica*

As a branch of zinc finger proteins, B-box proteins (BBXs) contain one/two B-box domain(s). Among plant BBXs, some members include an additional domain, CCT domain, at the C terminus, which proteins are designated Constans-like proteins (COLs) in some studies (Khanna et al. 2009). In apple, 17 COL proteins have been identified (Jin et al. 2017). To identify all the genes encoding BBX members with and without the CCT domain in the apple (*Malus domestica* Borkh.) genome, a local BLAST (blastp) was performed in the database of Genome Database for *Rosaceae* (short for GDR, <https://www.rosaceae.org/>) using the sequences of the above 17 BBXs (COLs) as queries. The sequences of known BBXs from rice, *Arabidopsis* and tomato were also used as queries in the same database. Trying not to miss potential candidates and taking account of credible results in above blasting, the cut-off E value was set to 0.001 as in previous studies (Li et al. 2011; Cui et al. 2015; Xu et al. 2016). The hidden Markov model (HMM) profile (Pfam00643)

Table 1 The detailed information of *MdBBX* members

Gene names	Gene ID from GDR database	Chromosome: location	CDS/bp	Size/AA	pI	MW
MdBBX1	MDP0000202669	chr1: 26862700...26863811	1023	340	5.74	38.14
MdBBX2	MDP0000294359	chr2: 19905176...19906517	1224	407	8.11	44.09
MdBBX3	MDP0000172036	chr3: 28997567...28999608	450	149	5.35	16.32
MdBBX4	MDP0000289278	chr5: 2765013...2765447	435	144	4.58	15.88
MdBBX5	MDP0000939920	chr5: 2779394...2779828	435	144	4.58	15.88
MdBBX6	MDP0000761905	chr5: 2792867...2793301	435	144	4.58	15.88
MdBBX7	MDP0000598183	chr5: 5786734...5786946	213	70	7.50	7.68
MdBBX8	MDP0000280947	chr5: 20737133...20745836	2577	858	6.13	94.70
MdBBX9	MDP0000140460	chr5: 20744637...20745869	546	181	8.41	20.25
MdBBX10	MDP0000733075	chr6: 16920878...16921438	561	186	4.55	20.68
MdBBX11	MDP0000271388	chr6: 22904175...22906599	1341	446	5.21	49.54
MdBBX12	MDP0000283949	chr7: 11349283...11353918	1458	485	7.14	53.05
MdBBX13	MDP0000264228	chr7: 25836126...25837798	1374	457	6.45	49.79
MdBBX14	MDP0000264845	chr7: 25837817...25839485	1443	480	8.44	52.27
MdBBX15	MDP0000551876	chr8: 1381940...1386799	1335	444	6.28	49.06
MdBBX16	MDP0000297093	chr9: 4850373...4852477	1179	392	5.76	43.56
MdBBX17	MDP0000194889	chr9: 6486783...6487555	693	230	4.37	24.71
MdBBX18	MDP0000198531	chr9: 6495007...6495595	558	185	4.11	20.07
MdBBX19	MDP0000232355	chr9: 6833588...6836630	1371	456	5.70	49.64
MdBBX20	MDP0000177126	chr9: 9619025...9620952	1248	415	6.14	45.38
MdBBX21	MDP0000587860	chr9: 12133412...12133600	189	62	6.68	6.74
MdBBX22	MDP0000298804	chr9: 24121267...24122742	897	298	4.93	31.64
MdBBX23	MDP0000222881	chr9: 24121313...24122788	990	329	5.28	35.22
MdBBX24	MDP0000800387	chr10: 10605557...10606492	627	208	6.44	23.26
MdBBX25	MDP0000901915	chr11: 33378342...33379421	1080	359	5.59	39.41
MdBBX26	MDP0000208320	chr11: 33997775...34000066	996	331	7.52	36.57
MdBBX27	MDP0000915501	chr12: 21541142...21542428	909	302	6.59	33.25
MdBBX28	MDP0000383112	chr13: 2085719...2086194	192	63	6.48	7.10
MdBBX29	MDP0000131980	chr13: 2544446...2546146	1365	454	5.22	50.48
MdBBX30	MDP0000314259	chr13: 2562817...2564517	1365	454	5.22	50.48
MdBBX31	MDP0000247810	chr13: 2563965...2564201	237	78	9.77	8.56
MdBBX32	MDP0000458656	chr13: 2569771...2570873	558	185	8.92	20.55
MdBBX33	MDP0000697407	chr13: 3060206...3061697	906	301	5.17	32.35
MdBBX34	MDP0000302297	chr13: 8847122...8862470	2772	923	5.94	101.43
MdBBX35	MDP0000565292	chr13: 15864540...15876017	1980	659	5.58	71.94
MdBBX36	MDP0000151848	chr13: 33621832...33624451	759	252	5.56	27.98
MdBBX37	MDP0000157816	chr14: 9991925...9992740	816	271	8.72	29.64
MdBBX38	MDP0000273201	chr14: 27661064...27667646	2226	741	4.77	83.53
MdBBX39	MDP0000298575	chr16: 774671...782658	2307	768	5.65	86.27
MdBBX40	MDP0000321380	chr16: 1224658...1229786	1752	583	6.03	64.67
MdBBX41	MDP0000185616	chr16: 1224811...1226412	1362	453	5.33	50.29
MdBBX42	MDP0000713113	chr16: 20659246...20667253	1404	467	6.97	50.37
MdBBX43	MDP0000140484	chr17: 5396728...5398846	1182	393	5.40	43.56
MdBBX44	MDP0000298635	chr17: 5396890...5398976	1182	393	5.40	43.56
MdBBX45	MDP0000321180	chr17: 6874441...6877832	945	314	4.90	34.65
MdBBX46	MDP0000198072	chr17: 6874660...6875470	729	242	4.53	26.37
MdBBX47	MDP0000782323	chr17: 9847401...9849311	1230	409	6.15	44.49
MdBBX48	MDP0000759984	chr17: 12792679...12793491	813	270	8.80	29.41
MdBBX49	MDP0000398010	chr17: 20057435...20058249	603	200	4.58	21.97
MdBBX50	MDP0000548690	chr17: 22895855...22899675	816	271	9.35	31.14

Table 1 (continued)

Gene names	Gene ID from GDR database	Chromosome: location	CDS/bp	Size/AA	pI	MW
MdBBX51	MDP0000664576	chr0: 12855188...12856983	732	243	4.95	26.65
MdBBX52	MDP0000128581	chr0: 66464234...66465313	1080	359	5.59	39.41
MdBBX53	MDP0000244238	chr0: 66464516...66465595	1080	359	5.59	39.41
MdBBX54	MDP0000232445	chr0: 34963592...34966330	900	299	5.00	32.92
MdBBX55	MDP0000128008	chr0: 7356965...7358489	609	202	5.88	22.76
MdBBX56	MDP0000321735	chr0: 12861348...12862789	564	187	7.49	20.48
MdBBX57	MDP0000686172	chr0: 64578669...64579103	435	144	4.58	15.89
MdBBX58	MDP0000272743	chr0: 74236542...74238208	1368	455	6.45	49.58
MdBBX59	MDP0000697109	chr0: 100056446...100056917	237	78	5.19	8.35
MdBBX60	MDP0000488955	chr0: 76682982...76683170	189	62	6.68	6.70
MdBBX61	MDP0000127949	chr0: 30937667...30939034	1062	353	9.19	38.78
MdBBX62	MDP0000313949	chr0: 63077159...63081560	1920	639	7.16	72.36
MdBBX63	MDP0000122414	chr0: 116968393...116968930	366	121	4.61	12.54
MdBBX64	MDP0000189746	chr0: 24652531...24654252	1371	456	8.83	50.29

Chr chromosome, *CDS* coding sequence, *AA* amino acid residues, *pI* isoelectric point, *MW* molecular weight

of the determined B-box domain from animals, plants and other organisms was also used to perform a global search of the apple proteome. Using the above strategies, approximately 800 acquired sequences of potential candidates were obtained. Each sequence was, respectively, submitted to the InterPro (<http://www.ebi.ac.uk/interpro/>) and SMART program (<http://smart.embl-heidelberg.de/>) to further examine the presence of B-box domain(s). Finally, 64 B-box containing proteins were identified and the encoding genes were named from *MdBBX1* to *MdBBX64*, based on their chromosomal locations. The detailed information of each *MdBBX* is presented in Table 1, including the gene name and gene ID from the GDR database, chromosome location, length of coding sequence and protein, theoretical isoelectric point, and molecular weight. One or two conservative B-box domains were included in all the *MdBBXs*, although the different *MdBBX* members varied significantly in size, ranging from 62 to 923 amino acids in length (Table 1, Fig. 1).

Motif logos of B-box domain(s) and phylogenetic analysis of the *MdBBX* family

The motifs logos of two B-box domains (B-box1 and B-box2) of *MdBBX* were constructed and are shown in Fig. 2. The distribution of conservative amino acid residues in these two domains was similar but not identical. There were five conservative Cys residues with four of them in the Cys-X-X-Cys motifs. The first Cys-X-X-Cys motif was at the beginning and the second in the middle of the B-box domain. The residues surrounding the second Cys-X-X-Cys motif were absolutely conservative, with Leu at the front and Asp following. There were also conservative residues, for example, Asp, Ala, and Asn and two His as indicated

in capital letters in Fig. 2. The sequences of B-boxes from apple, *Arabidopsis*, rice and tomato were compared (Fig. S2), which suggested that, in evolutionary relationship, the B-box of apple was closer with *Arabidopsis* than that with rice or tomato (Khanna et al. 2009; Huang et al. 2012; Chu et al. 2016).

To analyze the evolutionary relationship of *MdBBXs*, full-length amino acid sequences of 64 *MdBBXs* were used to construct a phylogenetic tree (Fig. 3). All 64 *MdBBXs* were divided into five groups and named as group I–V, according to the nomenclature used in *Arabidopsis* BBX family (Khanna et al. 2009). Members in group I, group II and group III contained an additional domain, CCT domain, with two B-boxes in groups I and II, and one B-box in group III. Although the members in both groups I and II contained two B-boxes, significant differences were observed in the amino acid sequences of B-box2; therefore, members fell into different groups (Fig. S3). The majority of members in group IV and group V had no CCT domain with two and one B-box domain(s), respectively. A few exceptions were found in the above groups, for example, *MdBBX7* and *MdBBX59* in group IV contained only one B-box rather than two B-boxes (they were presumptively classified to group V). However, multiple alignments showed that their B-box had more similarity with B-box1 in group IV than that in group V (Fig. S4). Individual *MdBBX* proteins might lose one B-box domain in a recent evolutionary event.

Chromosomal locations, gene duplication and gene structure analysis of *MdBBXs*

A total 50 of 64 *MdBBX* genes were successfully mapped to 15 different chromosomes (Fig. S5), and 14 *MdBBX* genes

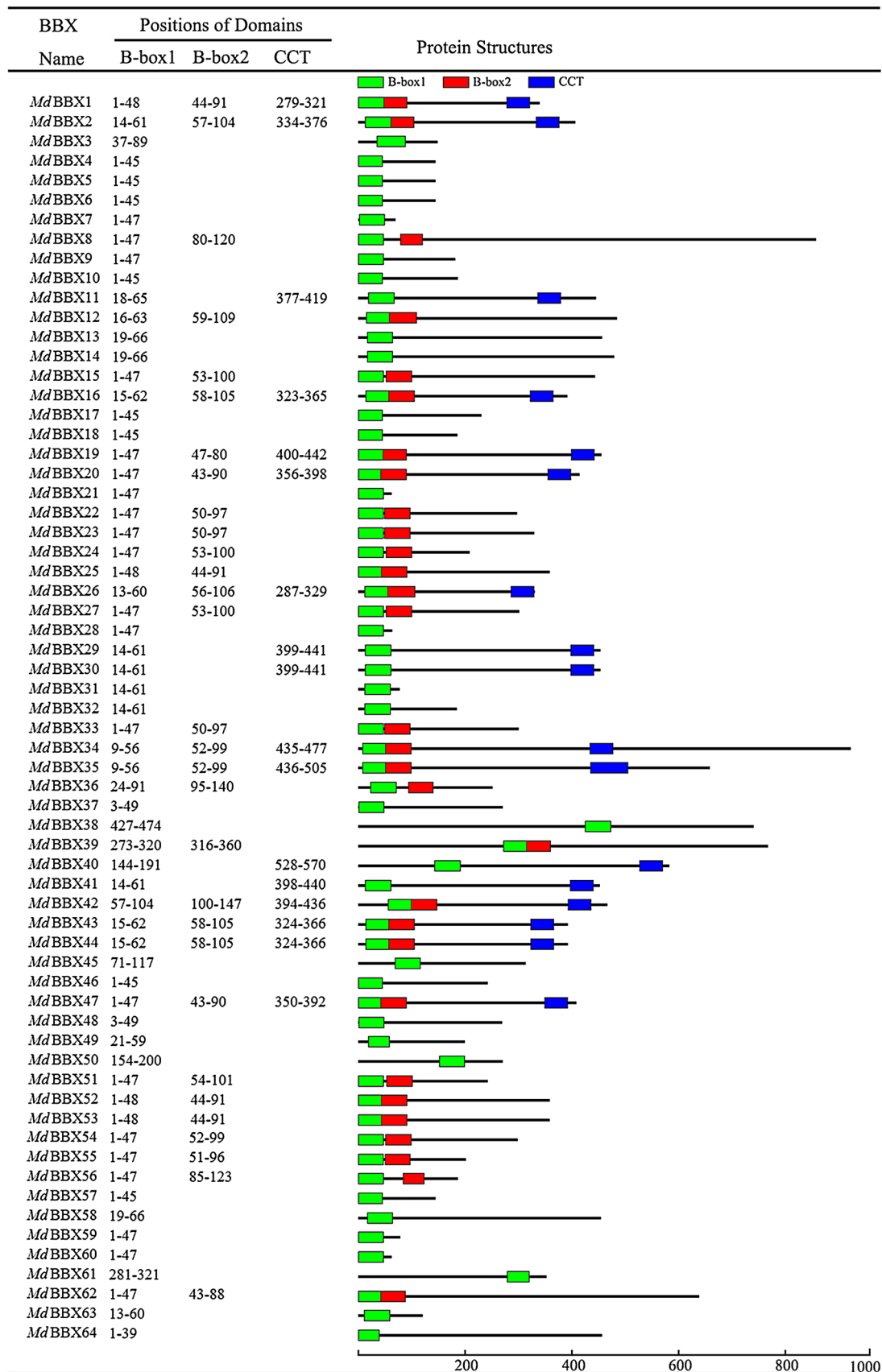


Fig. 1 Structures of the *Md*BBX proteins with the composition of the domains

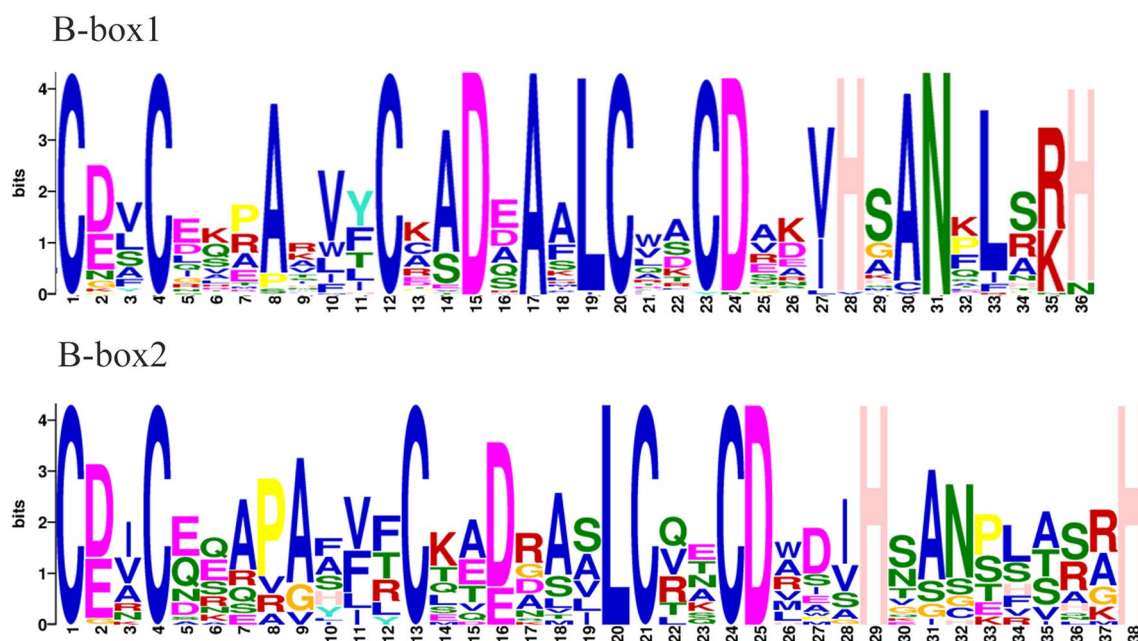


Fig. 2 The motifs of B-box 1 and B-box 2

could not be mapped in any chromosomes, which might be localized in the scaffolds of unsuccessful assembled genomic sequences (Li et al. 2011). Three chromosomes, i.e., chromosomes 9, 13 and 17, were present in more than eight *MdBBX* genes, in a cluster distribution. Some *MdBBX* genes in the clusters were formed by tandem duplication as their similarity was high than 70% in the same chromosome within 30 kb (marked in green color in Fig. S5). The genes with high similarity were also found in different chromosomes (Fig. S5), which means segmental duplication occurred. In addition, a genome-wide duplication event also occurred in apple and formed several intra-genomic homologous regions (Velasco et al. 2010). Large numbers of *MdBBX* genes were located in homologous regions (gray shadow region in Fig. S5) in different chromosomes, which suggested that genome duplication was another important way for expansion of *MdBBX* genes.

The ratios of Ka/Ks of were used to estimate the selection pressure of *MdBBXs* (Table S2). The majority of *MdBBX* paralogous pairs (19 of 22) were less than 0.8, which suggested that the *MdBBX* gene family has evolved mainly under the influence of strong purifying selection. Only two pairs of *MdBBX* genes had a ratio of Ka/Ks between 0.8 and 1, which indicates that some of *MdBBX* genes were under positive selection or relaxed selection for gene pairs (Wang et al. 2015). In addition, the intron–exon structure of encoding genes of *MdBBXs* also shared similar characteristics in different groups (Fig. S6).

The organ-specific expression patterns analysis of *MdBBXs*

To provide further clues to the putative roles of the *MdBBX* genes in apple development, the organ-specific expression patterns of *MdBBX* genes were investigated using the expression profiles from the GEO DataSets (GSE42873 from National Center for Biotechnology Information, <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE42873>). Gene expression patterns can provide important information for understanding the role of genes. According to the expression pattern in different organs, all 64 *MdBBX* members were divided into three groups, using the expression profiles from the GEO DataSets (GSE42873) (Fig. 4). Generally, *MdBBX* genes were widely expressed in flowers and fruits, indicating that *MdBBX* genes may play important roles during flowering regulation and fruits development. Differences of expression in different organs were found among different groups. Of 64 members, 11 genes were classified into Group A because they showed a higher expression in the leaves and flowers than that in other organs. In Group B, 32 genes were clustered together due to their highest expression in the flowers and the lowest expression in the seeds. In group C, 21 genes shared similar expression and were at high levels in young fruits, flowers, and leaves. The organ-specific expression patterns of *MdBBX* genes indicated that they might be directly or indirectly involved in the formation and development of reproductive organs, which was also confirmed in *Arabidopsis* BBXs (Putterill et al. 1995; Cheng and Wang 2005; Wenkel et al. 2006; Gangappa and Botto 2014).

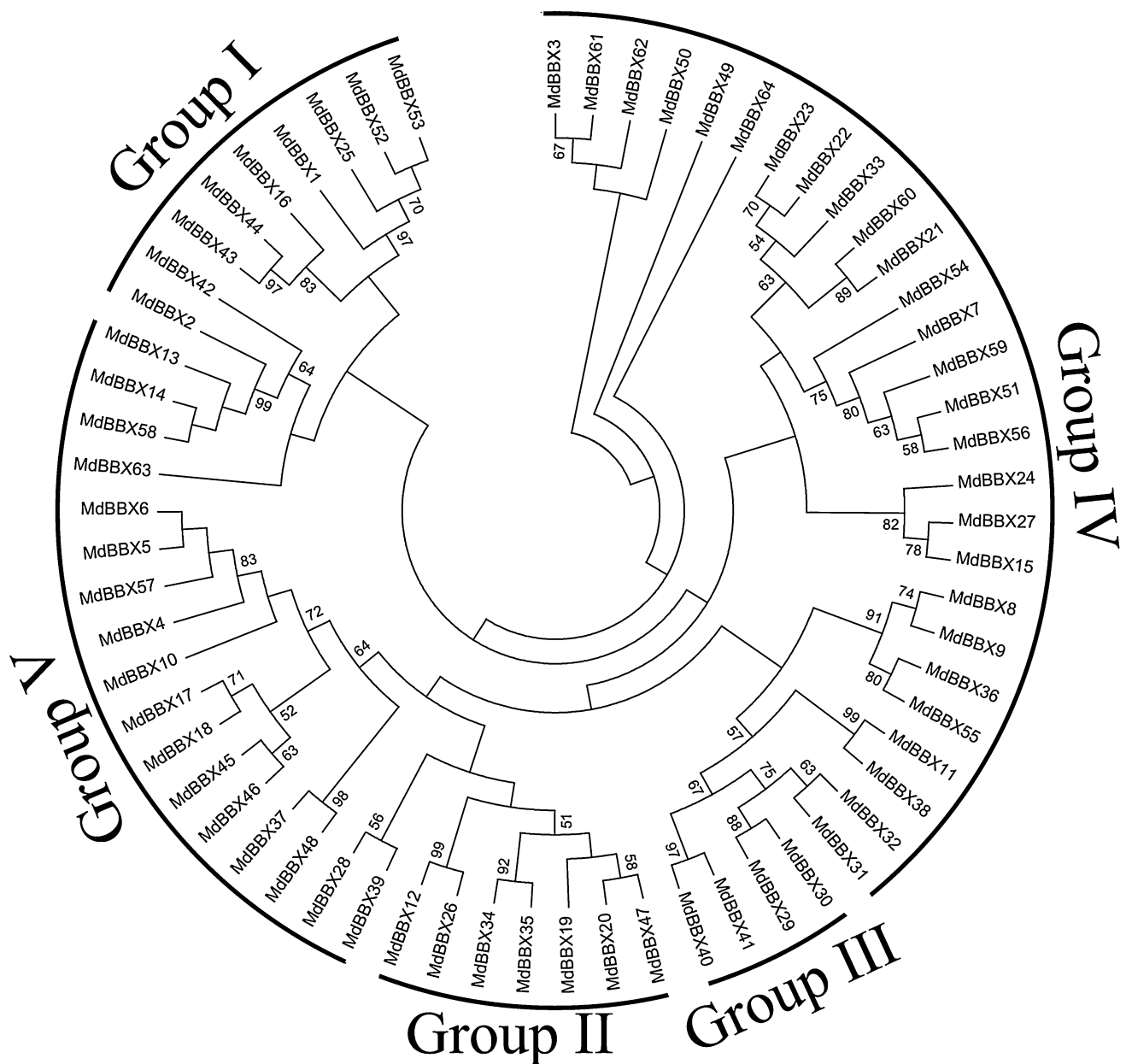


Fig. 3 Phylogenetic tree of the *MdBBX* family

Existence of stress-related *cis*-elements in the promoter of *MdBBXs*

Transcriptional control of stress-responsive genes is a crucial part of the plant response to a range of abiotic and biotic stresses. *Cis*-acting sequences in the promoter could regulate the expression of its following gene by binding with transcription factors (Singh et al. 2002). To identify putative *cis*-acting elements related to stresses, searches within the 1500 bp of the genomic sequence upstream of the *MdBBXs* 5'-UTRs were performed by submitting these promoter sequences to the PlantCARE database ([http://bioinformatics.](http://bioinformatics.psb.ugent.be/webtools/plantcare/html/)

[psb.ugent.be/webtools/plantcare/html/](http://bioinformatics.psb.ugent.be/webtools/plantcare/html/)) (Higo et al. 1999). Among 64 *MdBBXs* genes, a total of 48 promoter sequences could be found in the apple (*Malus domestica* Borkh.) genome database (which were downloaded from the database of GDR). The results showed that many *MdBBXs* genes harbored stress response-related *cis*-acting elements in their promoter regions (Table 2). More than 30 of 48 genes contained the MYB binding site (MBS), heat stress response elements (HSE), stress response elements (TC-rich repeats), anaerobic response elements (ARE) and salicylic acid response elements (TCA-element). More than 20 genes harbored stress response-related *cis*-acting elements, such

Table 2 The stress response-related *cis*-acting elements of *MdBBXs*

<i>Cis</i> -acting elements	Number of genes	Function of the <i>cis</i> -acting element
MBS	37	MYB binding site involved in drought-inducibility
HSE	36	<i>cis</i> -acting element involved in heat stress responsiveness
TC-rich repeats	35	<i>cis</i> -acting element involved in defense and stress responsiveness
ARE	34	<i>cis</i> -acting regulatory element essential for the anaerobic induction
TCA-element	31	<i>cis</i> -acting element involved in salicylic acid responsiveness
ABRE	29	<i>cis</i> -acting element involved in the abscisic acid responsiveness
TGACG-motif	25	<i>cis</i> -acting regulatory element involved in the methyl jasmonate responsiveness
CGTCA-motif	25	<i>cis</i> -acting regulatory element involved in the methyl jasmonate responsiveness
GARE-motif	23	Gibberellin-responsive element
LTR	22	<i>cis</i> -acting element involved in low-temperature responsiveness
ERE	21	Ethylene-responsive element

as ABA-response elements (ABRE), and low-temperature response elements (LTR). The presence of abiotic stress-responsive elements suggests that *MdBBXs* may be regulated by various stresses. Detailed information about types of *cis*-acting elements and the related genes are listed in Table S3.

Regulation of the expression of *MdBBX* genes by abiotic stresses

Based on the promoter analysis results, the expression of *MdBBXs* may be induced by abiotic stresses. To detect the quality of samples, one stress-sensitive gene (DREB gene: *MDP0000147009*) was selected as marker gene. Similar with the previous report, its expression was significantly up-regulated under different stresses both in leaves and roots (Fig. S7) (Zhao et al. 2012). Thus, to further investigate the potential functions of the *MdBBX* family, the expression of twelve randomly selected members was assayed under different stresses and exogenous ABA treatment. During PEG treatment, the majority of *MdBBXs* were up-regulated in roots and leaves (Fig. 5). Three of twelve *MdBBXs* in roots and four in leaves were up-regulated significantly, reaching approximately 40–65-fold in response to PEG treatment. Under high salinity and chilling stress, some *MdBBXs* were also up-regulated significantly in roots and leaves, for example, the expression of *MdBBX62* and *MdBBX19* increased nearly 20-fold in roots, respectively. During chilling stress, the expression of the gene *MdBBX50* was up-regulated both in roots and leaves, reaching more than ten- and 20-fold, respectively. In response to ABA treatment, the gene *MdBBX40* showed significantly up-regulated transcript levels both in roots and leaves, reaching more than five- and 25-fold, respectively. The expression of *MdBBX11*, *MdBBX27*, *MdBBX30* and *MdBBX54* in roots and *MdBBX62* in leaves also increased significantly. In general, most of twelve randomly selected *MdBBX* genes were sensitive to different abiotic stresses or to exogenous ABA treatment.

Discussion

Most transcription factors are specific to plants, information about transcription factors (including plant BBXs) are updated in PlantTFDB 4.0 (Jin et al. 2017). In this study, 64 genes encoding B-box proteins were identified in apple, which was much more than that from other plants, i.e., 32 for *Arabidopsis*, 30 for rice, 29 for tomato, 30 for potato and 37 for pear (Khanna et al. 2009; Huang et al. 2012; Chu et al. 2016; Talar et al. 2017; Zou et al. 2017). Most *MdBBX* genes were tandem duplication genes (as marked in green color in Fig. S5) or were located in homologous regions (indicated in the gray shadow region in Fig. S5), suggesting that gene tandem duplication, segmental duplication and genome-wide duplication in apple may lead to more BBX members in apple than those of other plants.

Phylogenetic analysis showed that all *MdBBXs* were divided into five groups (Fig. 3), which was similar to those of *Arabidopsis* and rice BBXs (Khanna et al. 2009; Huang et al. 2012). *MdBBX* members in groups I, II and IV contained two B-box domains. Different from the animal BBXs with two different types of B-boxes, the conservative amino acid sequences of two B-boxes in *MdBBXs* were similar, although they were not identical (Massiah et al. 2006, 2007). The early *MdBBXs* might originally have only one B-box domain (group V), and the other B-box domain was evolved in some cases during a duplication event (group IV), as most algae have a single B-box domain. An additional CCT domain was later generated in BBX proteins with double B-boxes (groups I and II), only differences existed in B-box2 in the two groups. In some cases, B-box2 disappeared and these members were classified into groups. The likely evolution of *MdBBXs* also occurred in *Arabidopsis* (Crocco and Botto 2013).

The CCT domain containing rich basic amino acids, always emerging with B-box domain(s), was only found in plant BBXs. Some CCT-containing BBXs have been

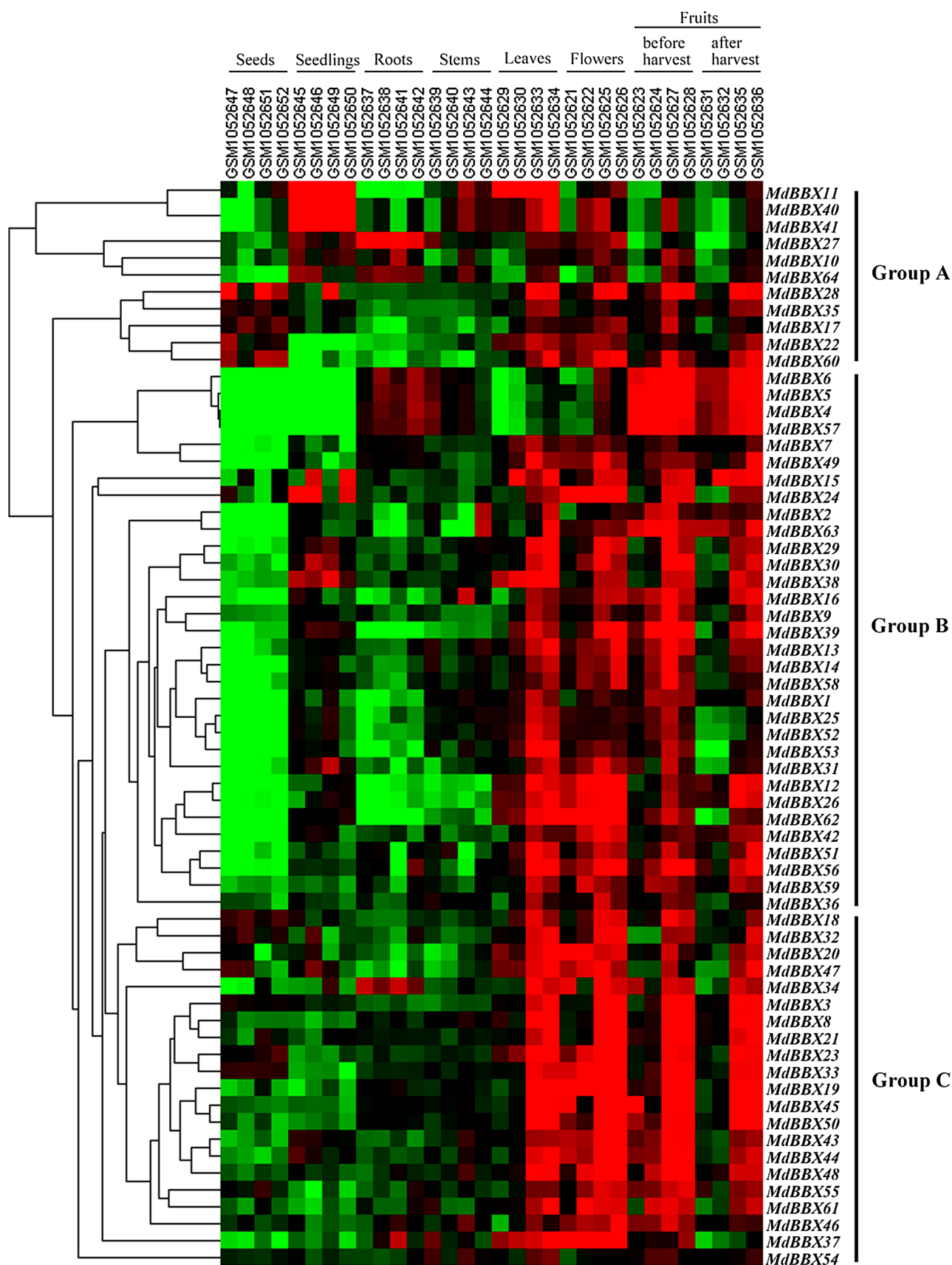


Fig. 4 Organ-specific expression pattern of *MdBBX*s

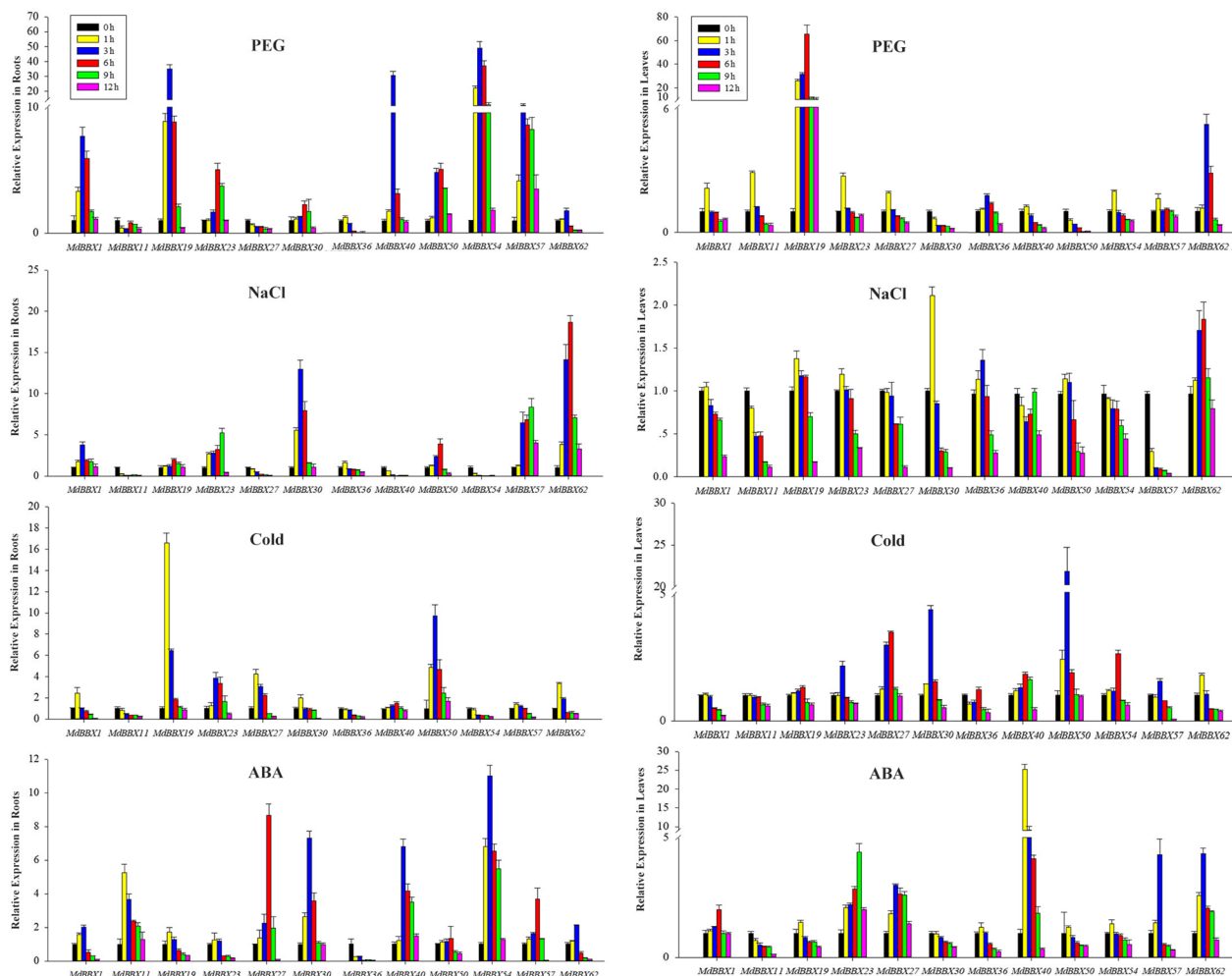


Fig. 5 The expression of *MdBBXs* in leaves and roots under different stresses. Data were normalized to the expression level of the actin gene. The mean value was calculated from three independent replicates. Vertical bars indicate the standard error of the mean

demonstrated to play roles in the regulation of flowering, such as *AtBBX1*, *OsBBX1* and *BvCOL1* (Putterill et al. 1995; Yano et al. 2000; Suarez-Lopez et al. 2001; Chia et al. 2008). Additionally, some have been associated with photomorphogenetic processes and abiotic stress responses (Nagao and Takano 2003; Datta et al. 2007; Fan et al. 2012). In this study, the analysis of *MdBBX* promoter regions revealed a series of *cis*-acting elements related to flowering, such as light-responsive elements (G-box, Sp1, Box I, Box 4) and circadian control elements (circadian). Flowering and photomorphogenesis belong to plant-specific processes; therefore, plant BBXs may have some functions different from those of animal BBXs.

The stress responses of plants are regulated by multiple signaling pathways, and there is a significant overlap between the patterns of gene expression that are induced by different stresses. Gene expression induced by stresses primarily occurred at the level of transcription. Several

closely related transcription factors have the potential to activate or repress genes through *cis*-acting sequences that respond to specific stresses. In this study, the analysis of *MdBBX* promoter regions revealed a series of frequent occurrence of *cis*-acting elements, such as MBS, HSE, ABRE or LTR. These elements basically participate in drought, high temperature, exogenous ABA responses and low temperature. A total of 48 *MdBBX* genes contained at least one of the above *cis*-elements, suggesting that these genes have a putative role in the response to abiotic and biotic stresses. Hormone-responsive *cis*-elements (TCA-element, ABRE, GARE and ERE) have been identified from the promoters of 31 *MdBBX* genes, so that the expression of these genes may be regulated by hormones.

Our results showed that the expression of 12 randomly selected *MdBBX* genes were induced to varying degrees under different treatments, including osmotic, high salt, cold stresses and exogenous ABA both in roots and leaves

(Fig. 5). In particular, the expressions were up-regulated by 20–40-fold under abiotic stresses or exogenous ABA treatment. Of the 12 selected *MdBBXs*, about half of the members contained a CCT domain, while other half of the members (*MdBBX23*, *MdBBX27*, *MdBBX36*, *MdBBX50*, *MdBBX54*, *MdBBX57* and *MdBBX62*) had no CCT domains in the N terminus of polypeptide chains. Whether they harbored a CCT domain, 12 selected *MdBBXs* responded to abiotic stresses. Therefore, we speculate that proteins containing a CCT domain or B-box domain may function in response to stress. When these two domains occur together, a synergistic effect may occur. In this study, we found that the *MdBBXs* were sensitive to different abiotic stresses or to exogenous ABA treatment. Our results suggested that plant BBX members may participate in responses to abiotic stress.

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Author contributions Designed the experiments: XYW and XL. Performed the experiments: XL, RL and YQD. Analyzed the data: XL, RL, YQD and XSC. Wrote the paper: XYW and XL. All authors read and approved the manuscript.

Compliance with ethical standards

Conflict of interest The authors declare no conflicts of interests.

Ethical approval This article does not contain any studies with animals performed by any of the authors.

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